Breno Livio Silva de Almeida

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Education

07.2024 – Present **PhD Computer Science, University of São Paulo and University of Leipzig**
Double degree PhD program with the University of São Paulo and University of Leipzig.
PhD Thesis: *Decoding the Language of Life: Towards the Limits of Biological Sequence Representation.*
Advisor (University of São Paulo): Prof. Dr. André C. P. L. F. de Carvalho 0000-0002-4765-6459
Advisor (University of Leipzig): Prof. Dr. Peter F. Stadler 0000-0002-5016-5191

01.2020 – 02.2024 **BSc Computer Science (5-Year Curriculum, Research-Intensive Track), University of São Paulo** with curricular specializations in Data Engineering and Data Science
GPA 9,4/10 (Rank: 7/116). Program completed on an accelerated timeline due to credit transfer from Computer Engineering. Thesis research conducted jointly between the Institute of Mathematics and Computer Sciences and the Helmholtz Centre for Environmental Research (UFZ), supported by a São Paulo Research Foundation (FAPESP) scholarship.
BSc Thesis: *BioDeepFuse: A Hybrid Deep Learning Approach with Handcrafted Features for Non-coding RNA Classification.* DOI: 10.1080/15476286.2024.2329451
Advisor: Prof. Dr. André C. P. L. F. de Carvalho 0000-0002-4765-6459

Funding Acquisition

To date, my research activities have been supported by **approximately € 112.944 in competitive national and international funding**, including major fellowships, sponsored training, and externally funded research projects. Full details are provided in the Honors, Awards, and Scholarships section.

Employment History

07.2025 – 06.2026 **Research Assistant,** Microbial Data Science group, Department of Computational Biology and Chemistry, Helmholtz Centre for Environmental Research (UFZ).

- Contributing to the EReMiD project for AI-assisted data type categorization and ontology alignment across different disciplines. More information: link
- Development of deep-learning approaches leveraging protein language models for functional annotation of metagenomic datasets through the AI MareExplore collaboration. More information: link

07.2022 – 09.2022 **Research Intern,** Big Data Biology Lab, Fudan University

- Large-scale metagenomic data analysis across multiple environments
- Rarefaction analysis of unique small open reading frames (smORFs)
- Co-author of a Nature Communications publication

05.2021 – 11.2022 **Teaching Assistant,** undergraduate level, 8 hours/week, Institute of Mathematics and Computer Sciences, University of São Paulo.

- Assisted in four undergraduate courses: Object-Oriented Analysis and Design, Statistics, Fundamentals of Data Science, and Computational Visualization
- Responsibilities included grading assignments, preparing exams, and delivering practical lectures to support student learning

Publication List

h-index: 4 **i10-index:** 3 **Citations:** 119 (Google Scholar), 74 (Scopus)

Google Scholar as of 2026-09-18 · Scopus as of 2026-09-18.

† *The authors contributed equally to the work.*

Preprints and Publications in Peer-Reviewed Journals

- **de Almeida, B. L.**, Bonidia, R. P., Bole, M., Avila-Santos, A. P., Stadler, P. F., Nunes da Rocha, U., & de Carvalho, A. C. P. L. F. (2026). BioAutoML-FAST: an automated machine-learning platform for reusable and benchmarked biological sequence models. *bioRxiv*. **DOI:** https://doi.org/10.64898/2026.04.18.719383. **Manuscript under revision at Bioinformatics following a major revision decision.**
- Duan, Y., Santos-Júnior, C. D., Schmidt, T. S., Fullam, A., **de Almeida, B. L.**, Zhu, C., ... & Coelho, L. P. (2024). A catalog of small proteins from the global microbiome. Nature Communications, 15(1), 7563. **DOI:** https://doi.org/10.1038/s41467-024-51894-6.
- **de Almeida, B. L.**†, Avila Santos, A. P.†, Bonidia, R. P., Stadler, P. F., Stefanic, P., Mandic-Mulec, I., ... & de Carvalho, A. C. (2024). BioDeepfuse: a hybrid deep learning approach with integrated feature extraction techniques for enhanced non-coding RNA classification. RNA biology, 21(1), 410-421. **DOI:** https://doi.org/10.1080/15476286.2024.2329451.
- Bonidia, R. P., Avila Santos, A. P., **de Almeida, B. L.**, Stadler, P. F., Nunes da Rocha, U., Sanches, D. S., & De Carvalho, A. C. (2022). Information theory for biological sequence classification: A novel feature extraction technique based on tsallis entropy. Entropy, 24(10), 1398. **DOI:** https://doi.org/10.3390/e24101398.
- Bonidia, R. P., Santos, A. P. A., **de Almeida, B. L.**, Stadler, P. F., da Rocha, U. N., Sanches, D. S., & de Carvalho, A. C. (2022). BioAutoML: automated feature engineering and metalearning to predict noncoding RNAs in bacteria. Briefings in Bioinformatics, 23(4), bbac218. **DOI:** https://doi.org/10.1093/bib/bbac218.
- Tavares, D., Barreto, M. C. A., **de Almeida, B. L. S.**, Florentino, B. R., Tetzner, F., Goedert, G., Struchiner, C., Bonidia, R., & de Carvalho, A. Dominique: An AI-powered fact-checking chatbot for democratizing access to reliable information. Oxford Open Digital Health. **Accepted for publication**.
- **de Almeida, B. L. S.**†, Bonidia, R.†, Florentino, B.†, Barreto, M. C. A.†, Constantino, J. L. R.†, Tavares da Silva, D.†, Zwicker, G. G., Rocha, U., Goedert, G., Struchiner, C., & de Carvalho, A. AutoAI Hub: From data to society, democratizing machine learning for everyone. Oxford Open Digital Health. **Accepted for publication**.

Conference Proceedings

- **de Almeida, B. L. S.**†, Queiroz, A. P.†, Santos, A. P. A.†, Bonidia, R. P., da Rocha, U. N., Sanches, D. S., & de Carvalho, A. C. P. D. L. F. (2021, November). Feature importance analysis of non-coding dna/rna sequences based on machine learning approaches. In Brazilian Symposium on Bioinformatics (pp. 81-92). Cham: Springer International Publishing. **DOI:** https://doi.org/10.1007/978-3-030-91814-9_8.

Academic Service

Journal Peer Review

09.2026 – Present **Ad hoc Reviewer, Journal of Thermal Biology** (Elsevier). One manuscript reviewed.

03.2025 – Present **Ad hoc Reviewer, Briefings in Bioinformatics** (Oxford University Press). One manuscript reviewed across two rounds.

Undergraduate Thesis Examination

Examination Committee Member. Served on six undergraduate thesis examination committees.

Honors, Awards, and Scholarships

2026 **HIDA-Sponsored Participant – The Carpentries Instructor Training**, selected for one of 15 annually sponsored training places offered by the Helmholtz Information & Data Science Academy (HIDA) to Helmholtz researchers. The program provides evidence-based training in scientific and data science instruction and leads toward certification as a Carpentries Instructor. Equivalent training value of USD 1.600 (approximately € 1.360; exchange rate 1 USD = 0,85 EUR).

2025 **Google PhD Fellowship in Health** awarded to support outstanding and innovative research in computer science and related fields, providing total funding of USD 30.000 over two years (approximately € 25.500; exchange rate 1 USD = 0,85 EUR).

2025 **ISME Scholar Mobility Fund** awarded with funding of € 2.300 for a research period in July 2026 at the Helmholtz Centre for Environmental Research (UFZ) in Leipzig, Germany.

2024 **J. F. Marar Artificial Intelligence Prize for Undergraduate Research** awarded with a prize of R$10.500 for the project *BioDeepFuse: Empowering Researchers in Life Sciences with Deep Learning* recognized for academic excellence, social impact, and alignment with the United Nations Sustainable Development Goals (SDGs) (approximately € 1.680; exchange rate 1 R$ = 0,16 EUR).

2024 **Prototypes for Humanity Travel Grant** awarded to BioAutoML, selected as a top 100 project among nearly 3.000 global submissions, with funding of USD 760 to support presentation at the international festival in Dubai (approximately € 646; exchange rate 1 USD = 0,85 EUR).

2024 **Doctoral Scholarship** awarded for the research project *"Decoding the language of life: Towards protein classification and design through large language models"*, funded by the São Paulo Research Foundation (FAPESP), grant #2024/10958-1. Total funding of R$453.168,00 (approximately €77.039; exchange rate 1 R$ = 0,17 EUR).

2024 **Technical Training Scholarship** awarded for the research project *"Democratization of machine learning in life sciences and annotation of protein families"*, funded by the São Paulo Research Foundation (FAPESP), grant #2024/03304-5. Total funding of R$10.553,20 (approximately € 1.688; exchange rate 1 R$ = 0,16 EUR).

2023 **Scientific Initiation Scholarship** awarded for the research project *"Machine Learning applied to multiomics analysis for exploration of microbial communities"*, funded by the São Paulo Research Foundation (FAPESP), grant #2023/00264-0. Total funding of R$7.920,99 (approximately € 1.267; exchange rate 1 R$ = 0,16 EUR).

2023 **Scientific Initiation Scholarship** awarded for the research project *"Automated machine learning for life sciences"*, funded by the National Council for Scientific and Technological Development (CNPq), grant #101243/2023-5. Total funding of R$1.500 (approximately € 240; exchange rate 1 R$ = 0,16 EUR).

2021 – 2022 **Scientific Initiation Scholarship** awarded for the research project *"Classification and Analysis of Bacterial Non-coding RNA Sequences Using Machine Learning Techniques"*, funded by the São Paulo Research Foundation (FAPESP), grant #2021/08561-8. Total funding of R$7.652,70 (approximately € 1.224; exchange rate 1 R$ = 0,16 EUR).

Skills

Languages — Portuguese (Native), English (C1, Full Professional Proficiency), German (A1.2, Elementary Proficiency).

Programming — Python, R, Bash, Nix, Java, C, C++, SQL.

Machine Learning & Data — Scikit-learn, TensorFlow, Pandas, Polars, NumPy, SciPy, NetworkX.

Visualization & Apps — Matplotlib, Plotly, Streamlit.

Software & Systems — Jupyter Notebook, R Markdown, Git, Linux, LaTeX.

Bioinformatics — Biopython, BLAST, Prodigal, HMMER, EMBOSS, Infernal, seqkit, CD-HIT.

Invited Talks and Selected Oral Presentations

- **de Almeida, B. L.** Da bancada ao teclado: ferramentas gratuitas para a pesquisa científica [From bench to keyboard: free tools for scientific research]. **Invited talk**, 36th FisFar Summer Course, Federal University of Minas Gerais (UFMG), Brazil. January 2027. *Scheduled.*

- **de Almeida, B. L.**, Kamath, S., Santos, A. P. A., Bonidia, R. P., Espíndola-Hernández, P., Stadler, P. F., Wierckx, N., Borchert, E., Rother, D., Oliveira, A., de Carvalho, A. C. P. L. F., & Rocha, U. N. Combining protein language models with different architectures expands protein family annotation 14.2-fold across 16 million open reading frames from 3,802 environmental metagenomes. In: 20th International Symposium on Microbial Ecology (ISME20), 2026, Auckland, New Zealand.

National and International Collaborations

- **Auto AI-Pandemics: Democratizing Machine Learning for the Analysis, Study, and Control of Epidemics and Pandemics (2024 – Present)**, research initiative funded by the International Development Research Centre (IDRC) under the *AI for Global Health* program, in partnership with UK International Development, with total funding of $362,500 CAD. The project involves a multidisciplinary collaboration among the University of São Paulo, Fundação Getulio Vargas, University of Brasília (UnB), and Federal University of Technology – Paraná (UTFPR) in Brazil; the University of Leipzig and the Helmholtz Centre for Environmental Research (UFZ) in Germany; and Queen's University and the London School of Hygiene and Tropical Medicine in the United Kingdom.
- **NFDI4Microbiota (2024 – Present)**, consortium of the German National Research Data Infrastructure (NFDI) dedicated to the sustainable management, standardization, and sharing of microbiological and microbiome-related research data. The initiative serves as the central national hub for microbiology research data in Germany, supporting the community with FAIR, open, and AI-ready data, metadata standards, analysis services, and training. The consortium is driven by 10 core partner institutions and supported by over 50 participating organizations, encompassing leading universities and non-university research centers, and contributes to the broader One NFDI vision by developing interconnected, cross-disciplinary data infrastructures that accelerate data-driven discovery in microbiology.
- **AI MareExplore (2025 – Present)**, interdisciplinary research initiative focused on the development of AI-driven microbial bioprospecting pipelines for the sustainable exploitation of marine microbial biocatalysts toward a blue and circular bioeconomy. The project leverages large-scale public marine genomic datasets to train AI language models for biocatalyst prediction across multiple application domains, including carbon capture and plastic waste revalorization. Participating centers include the Helmholtz Centre for Environmental Research (UFZ), GEOMAR Helmholtz Centre for Ocean Research Kiel, Forschungszentrum Jülich (FZJ), and GFZ Helmholtz Centre for Geosciences, with strong integration into international, EU, and national research projects ensuring feasibility, scalability, and dissemination.

Other Academic Activities

Teaching

- **Institute of Mathematics and Computer Sciences, University of São Paulo**. São Carlos, Brazil. Teaching assistant for the following classes: Object-Oriented Analysis and Design, Statistics, Fundamentals of Data Science, and Computational Visualization (05.2021 – 11.2022).
- **17th Symposium on Bacterial Genetics and Ecology (BAGECO 2025)**. Graz, Austria. Taught in the Workshop Metagenomics with classes on "Experiment design and Do's and Don'ts on computer experiments" and "From Sequences to Predictions – End-to-end analysis using BioAutoML" (06.2025).
- **Summer School in Trends in multi-omics data analysis for Microbial Ecology and Biotechnology**. Leipzig, Germany. Taught classes on Unix/Bash/Git, machine learning, alignment-based annotation and alignment-free annotation (07.2025 and 07.2026).
- **13th International Symposium on Anaerobic Microbiology (ISAM)**. Leipzig, Germany. Taught in the NFDI4Microbiota bioinformatics workshop with a class on "The first step to make your data useful: (meta)data submission to public repositories" (09.2025).

Complementary education

- **Summer School in Trends in multi-omics data analysis for Microbial Ecology and Biotechnology**, Helmholtz Centre for Environmental Research (UFZ), Leipzig, Germany (07.2025).
- **15th Summer Course in Bioinformatics**: Faculty of Medicine of the University of São Paulo (FMUSP), São Paulo, Brazil (01.2024).
- **Applied Data Science with Python Specialization**, University of Michigan, Coursera, Online (2020 – 2021).
- **Data Science Specialization**, Johns Hopkins University, Coursera, Online (2020 – 2021).
- **NUTR104x: Nutrition and Health: Human Microbiome**, Wageningen University, WUR, edX, Online (2020).

Extracurricular Activities

- **Co-founder and coordinator**, efestus, Institute of Mathematics and Computer Sciences, University of São Paulo (06.2018 – 02.2021). Student organization focused on computer science and data science for social good, especially on debating the ethical implications of technological advances. Fully funded participation, including travel and U.S. visa costs, at the TechShift Summit, Chapel Hill, NC (USA)
- **Software division member**, Warthog Robotics, Institute of Mathematics and Computer Sciences, University of São Paulo (06.2017 – 12.2019). Participation in events such as the Latin American and Brazilian Robotics Competition (LARC). Taught Git and Object-Oriented Programming to new members.

Selected Abstract Presentations

Full list available at: brenoslivio.github.io/works/

- **de Almeida, B. L.**, Bonidia, R. P., Bole, M., Santos, A. P. A., Stadler, P. F., Rocha, U. N., & de Carvalho, A. C. P. L. F. From sequences to traits: An AI-driven AutoML framework for genome-resolved microbial ecology. In: 20th International Symposium on Microbial Ecology (ISME20), 2026, Auckland, New Zealand.
- Bonidia, R. P., Florentino, B. R., Barreto, M. C. A., **de Almeida, B. L.**, Constantino, J. L. R., Silva, D. T., Rocha, U. N., Goedert, G. T., Struchiner, C. J., & de Carvalho, A. C. P. L. F. AutoAI Hub: From Data to Society, Democratizing Machine Learning for Everyone. In: Human Genome Meeting 2026, 2026, Athens, Greece.
- Santos, A. P. A., **de Almeida, B. L.**, Bole, M., Kamath, S., & Rocha, U. N. A FAIR-by-Design and Standard-Compliant System for Automated Submission of Microbial Genomes to the European Nucleotide Archive. In: VAAM Annual Conference, 2026, Berlin, Germany.
- **de Almeida, B. L.**, Kamath, S., Santos, A. P. A., Bonidia, R. P., Espíndola-Hernández, P., Stadler, P. F., Oliveira, A., de Carvalho, A. C. P. L. F., & Rocha, U. N. Exploration of multiple protein language models and over 16 million open reading frames recovered from thousands of terrestrial metagenomes uncover protein domains and families in 13,486 previously unknown orthologs. In: 17th Symposium on Genetics and Bacterial Ecology (BAGECO 2025), 2025, Graz, Austria.
- Bonidia, R. P., Santos, A. P. A., **de Almeida, B. L.**, Stadler, P. F., Rocha, U. N., Sanches, D. S., & de Carvalho, A. C. P. L. F. BioAutoML: End-to-End Machine Learning Package for Life Sciences. In: 10th FEMS Congress of European Microbiologists, 2023, Hamburg, Germany.

Professional Associations

- **International Society of Microbial Ecology (ISME)**, since 10.2025
- **Federation of European Microbiological Societies (FEMS)**, since 11.2025